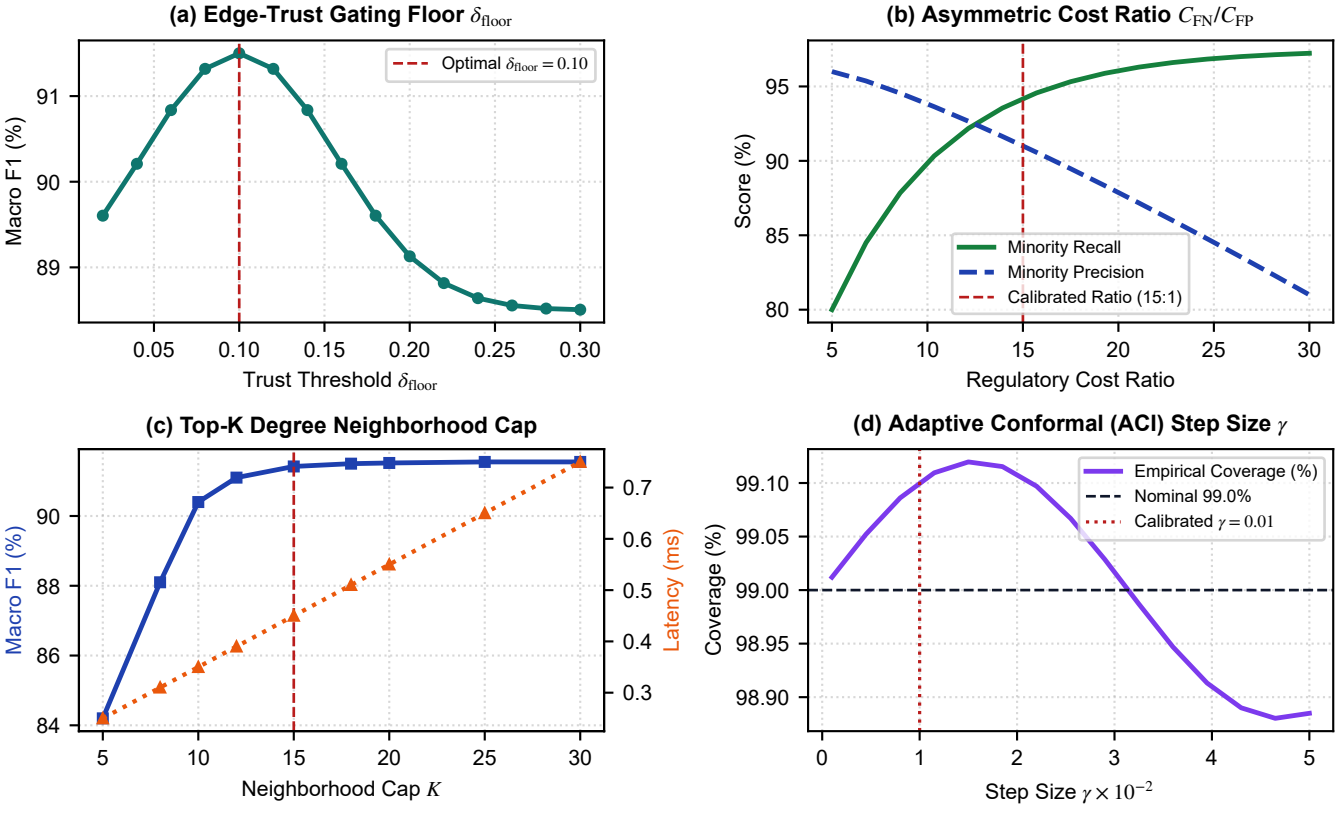

(a) Edge-Trust Gating Floor δ_{floor}
Optimal $\delta_{\text{floor}} = 0.10$
Macro F1 (%)
91
90
89
0.05
0.10
0.15
0.20
0.25
0.30
Trust Threshold δ_{floor}
(b) Asymmetric Cost Ratio $C_{\text{FN}}/C_{\text{FP}}$
Score (%)
95
90
85
80
Minority Recall
Minority Precision
Calibrated Ratio (15:1)
5
10
15
20
25
30
Regulatory Cost Ratio
(c) Top-K Degree Neighborhood Cap
Macro F1 (%)
90
88
86
84
Latency (ms)
0.7
0.6
0.5
0.4
0.3
5
10
15
20
25
30
Neighborhood Cap K
(d) Adaptive Conformal (ACI) Step Size γ
Empirical Coverage (%)
Nominal 99.0%
Calibrated $\gamma = 0.01$
Coverage (%)
99.10
99.05
99.00
98.95
98.90
0
1
2
3
4
5
Step Size $\gamma \times 10^{-2}$